



Light-Activated ROBOT

MANMOHAN SINGH, VIVEK GOWDA, RAMIKSHA C SHETTY, AND SAHANA P

The mini light-activated robot proposed here works in ambient light. It employs sensors to detect variations in light intensity, which allows it to navigate and perform tasks in diverse environments. This mini-robot exploration combines robotics and light sensing technology, showcasing the potential for autonomous systems to adapt to changing light conditions. The block diagram of the light-activated robot is shown in Fig. 1.

This mini-robot introduces a captivating exploration into the realm of robotics by combining the light-dependent resistor (LDR) and the 555 timer IC. The core concept involves creating a light-activated robot, where the LDR functions as the sensory 'eyes,' and the 555 timer serves as the controlling 'brain.' As light intensity changes, the resistance values of the LDR detect these variations, prompting the 555 timer to orchestrate the robot's movements. The result is a robot that dynamically responds to its environment, moving in coordination with the presence or absence of light. This hands-on endeavour provides a foundational understanding of sensors, timers, and motor control, bridging the gap between theory and practical application. The author's prototype on a breadboard is shown in Fig. 2.

Circuit and working

Fig. 2 shows the circuit diagram of the light-activated robot. It is built around two 555 timers (IC1, IC2), two DC motors (M1, M2), two light-dependent resistors (LDR1, LDR2), and a few other compo-

nents. For the power supply, a 9V power adaptor or a 9V battery may be used.

The idea behind the light-activated robot is to create a versatile and adaptive robotic system that can respond to changes in lighting conditions. Light sensors are utilised to enable the robot to perceive its

surroundings and make decisions accordingly.

The circuit comprises two divisions: one is LDR, the light-sensing element, and the other is the 555 activator. The LDR1 is paired with resistor R1 to form a potential divider to feed the trigger voltage to the IC 555 (IC1). It gives maximum voltage when no light is incident on it and minimum voltage when light falls on the LDR.

The resistance of the LDR is inversely proportional to the intensity of light falling on it. It implies that if the intensity of incident light is high, the resistance of the LDR will be less, and vice versa.

The activator is built around a 555 IC, which is wired as an astable multivibrator. This principle is used here to make the 555 act as a switch when light is incident on the LDR.

The working of this robot is

simple. Timer IC 555 gets activated when its discharge pin 7 receives a voltage greater than 0.8V. Once the IC is activated, the voltage at pins 2 and 6 needs to be between 1/3rd and 2/3rd of the supply voltage for the output to be high. For example, if the voltage at the discharge pin 7 is above 0.8V, and the voltage at pins 2 and 6 is half the supply voltage, the output goes high to turn the motor.

In the circuit, a voltage divider is created using the LDR and a resistor. It is then con-

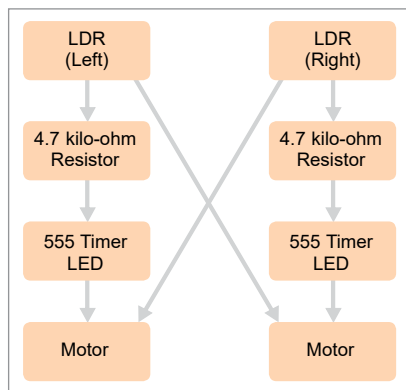


Fig. 1: Block diagram of light-activated robot

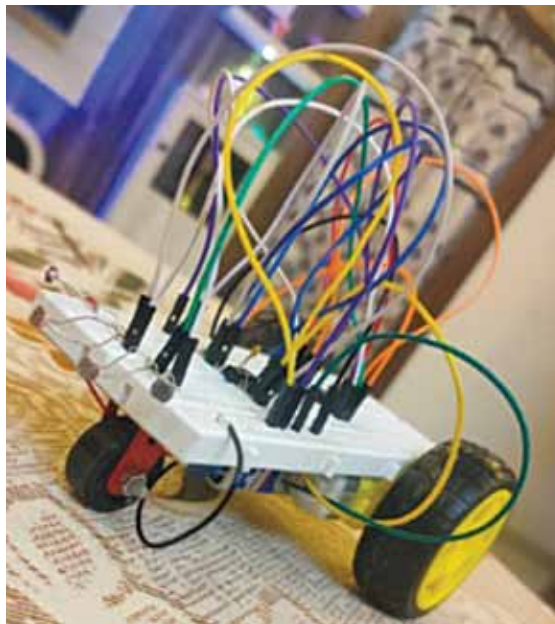


Fig. 2: Authors' prototype